

★ An experiment is a process whose outcome is not known with certainty.

- The set of all possible outcomes of an experiment is called sample space (denoted by S)

- The outcomes themselves are called sample points in the sample space.

$$S = \{H, T\}$$

$$S = \{1, 2, 3, 4, 5, 6\}$$

Sample points $\rightarrow$ Random variables

Range of Random Variables : $-\infty$ to $+\infty$

$$S = \{1.7, 1.8, \dots, 2.0\}$$

★ Discrete : A random variable X is said to be discrete if it can take on a countable number of values.

- The probability that X takes on the value x_i is given by:

$$P(x_i) = P(X = x_i) \text{ for } i = 1, 2, 3, \dots$$

- Here, we must have

$$\sum_{i=1}^{\infty} P(x_i) = 1$$

as the probability mass function

probability

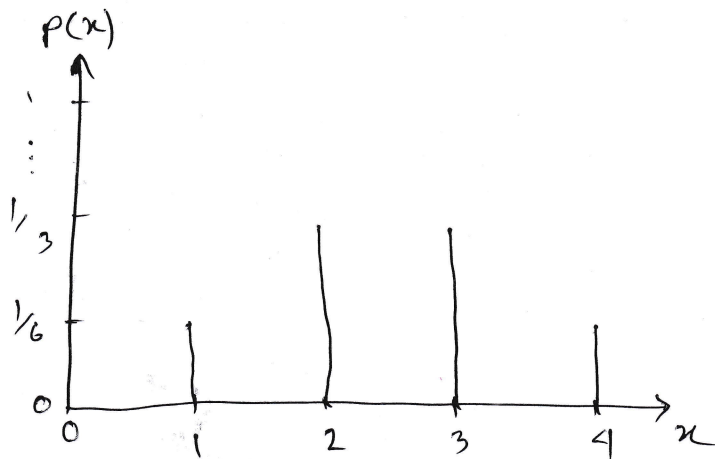
The distribution function $F(x)$ of the random variable X is defined as follows.

- $F(x) = P(X \leq x)$ for $-\infty < x < +\infty$
- Thus, $F(x)$ is the probability that, when the experiment is done, the random variable X will take on a value no larger than x .
- $F(x)$ has the following properties:
 1. $0 \leq F(x) \leq 1$ for all x .
 2. $F(x)$ is nondecreasing.
if $x_1 < x_2$, then $F(x_1) \leq F(x_2)$
 3. $\lim_{x \rightarrow \infty} F(x) = 1$ and $\lim_{x \rightarrow -\infty} F(x) = 0$
since X takes only finite values

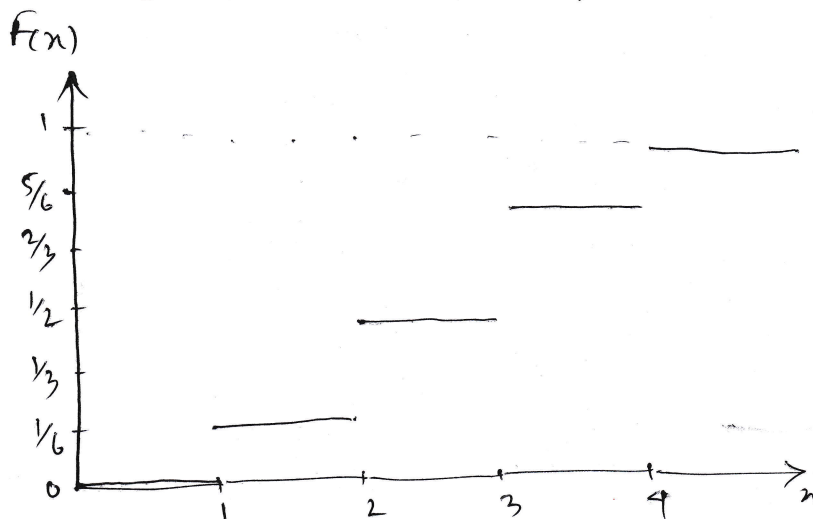
For the inventory example :

Size of demand (x)	Probabilities
1	$\frac{1}{6}$
2	$\frac{1}{3}$
3	$\frac{1}{3}$
4	$\frac{1}{6}$

Probability mass function :



Probability distribution function :



Continuous: A random variable X is said to be continuous if there exists a nonnegative function $f(x)$ such that for any set of real numbers B :

$$P(X \in B) = \int_B f(x) dx \quad \text{and} \quad \int_{-\infty}^{+\infty} f(x) dx = 1$$

For any real number x ,

$$P(X = x) = P(X \in [x, x])$$

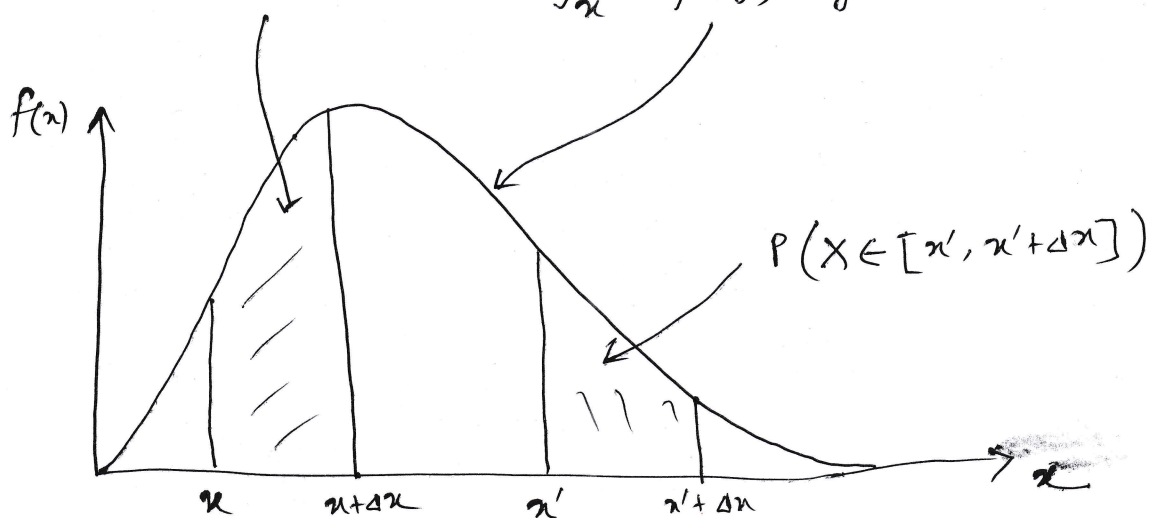
$$= \int_x^x f(y) dy$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(y_i) \Delta y \quad \Delta y = \frac{x - x}{n} = 0$$

$$= 0$$

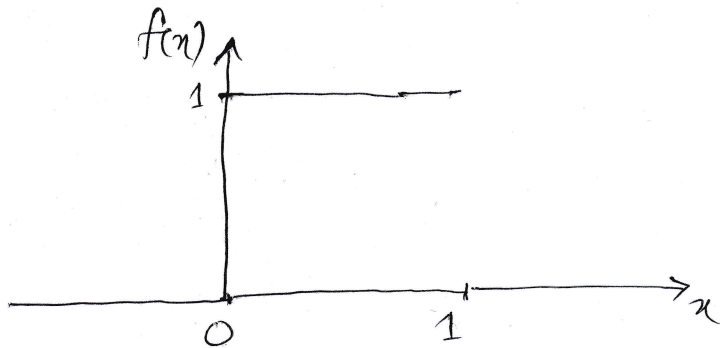
When $\Delta x > 0$,

$$P(X \in [x, x + \Delta x]) = \int_x^{x + \Delta x} f(y) dy \quad \text{--- (1)}$$



For uniform random variable on the interval $[0, 1]$ we have the following probability density function:

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$



Now, we have the following probability distribution function:

$$F(x) = \int_0^x f(y) dy$$

$$= \int_0^x 1 dy$$

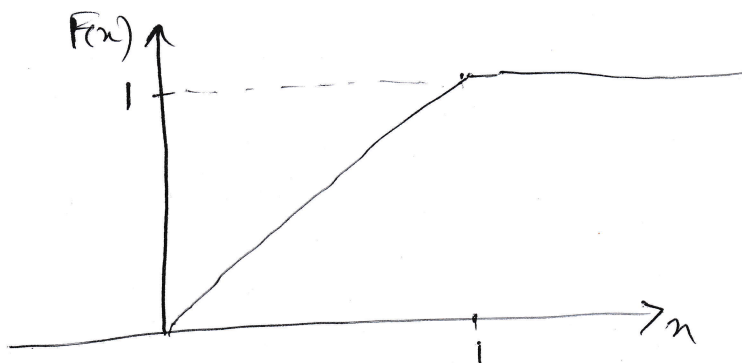
$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n 1 \cdot \Delta y$$

$$= \lim_{n \rightarrow \infty} x \cdot \frac{x}{x}$$

$$= x$$

$$\Delta y = \frac{x-0}{n} = \frac{x}{n}$$

$$\sum_{i=1}^n 1 = n$$



From Eqn. ①,

$$P(X \in [x, x+\Delta x]) = \int_x^{x+\Delta x} f(y) dy$$

When $0 \leq x < x+\Delta x \leq 1$,

$$\int_x^{x+\Delta x} f(y) dy = \int_x^{x+\Delta x} 1 \cdot dy$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n 1 \cdot \Delta y$$

$$= \lim_{n \rightarrow \infty} n \cdot \frac{\Delta x}{n}$$

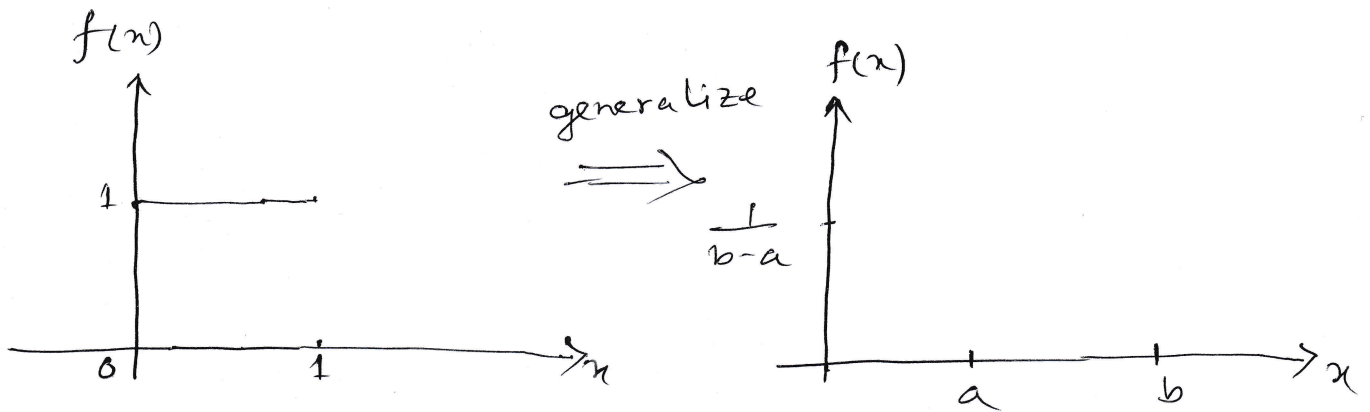
$$= \Delta x$$

$$\Delta y = \frac{x+\Delta x - x}{n} = \frac{\Delta x}{n}$$

$$\sum_{i=1}^n 1 = n$$

It indicates that a uniform random variable on $[0,1]$ is equally likely to fall in any interval of length Δx between 0 and 1, which justifies the name $\rightarrow$ "uniform".

Uniform Distribution: has constant probability within a certain domain.



$$f(x) = 1$$

$$= \text{height} \times \text{width}$$

$$= \text{height} \times (b-a)$$

$$\text{height} \times (b-a) = 1$$

$$\therefore \text{height} = \frac{1}{b-a}$$

Properties $\Rightarrow$ Range: $[a, b]$

$$\text{Mean } (\mu): \frac{b+a}{2}$$

$$\text{Variance: } \frac{(b-a)^2}{12}$$